Spline Tool

TrenchBroom | User Guide

The Spline Tool sweeps a template brush (or a group of brushes) along a smooth curve, producing solid, grid-aligned geometry that bends and twists to follow the path. Use it for pipes, rails, tunnels, cables, arches, tracks — anything that repeats a cross-section along a line.

Getting Started

- **Activate the tool** from the toolbar (the curve icon) or the **Edit** → **Tools** → **Spline Tool** menu. The default shortcut is **Y**.
- When the tool is active, **every spline in the map is shown** and can be picked up for editing. Click a spline's points or its generated brushes to start editing that spline.
- The tool's controls appear in the **tool bar strip** below the main toolbar.

Building a spline

- Press **Add** to enter add-point mode, then click in a viewport to drop control points. Each click appends a point; a preview line shows where the next point will connect.
- Add mode is **always off when you switch to the tool**, so you never create stray points by accident. Toggle **Add** off (or press **Esc**) to stop adding and go back to selecting points.
- You need at least **two points** and a linked template before any brushes appear.

Editing Control Points

- **Select** a point by clicking it (with Add mode off). The selected point's Roll, Scale and Locked values are shown in the tool bar.
- **Drag** a point to move it. Hold **Alt** to move vertically (3D view), and **Ctrl** to toggle grid snapping — exactly like moving a brush.
- **Arrow keys** move the selected point by one grid step; **Page Up / Page Down** move it on the vertical axis.
- **Insert between points:** select a point that sits between two others, and the next point you add is inserted right after it instead of at the end — handy for refining a section of the curve.
- **Remove** deletes the selected point (or the last one if none is selected).
- **Double-click** a point to toggle its Locked state quickly.

Per-point shape

Control	What it does
Roll	Twists the swept cross-section around the curve at this point, in degrees. The twist blends smoothly between points.
Scale	Scales the cross-section at this point. Values taper the profile between points (e.g. 1.0 → 0.5 makes a cone-like narrowing).
Locked	Anchors the sweep's orientation at this point. A twist introduced by rolling other points cannot propagate past a locked point, so lock a point to “pin” the geometry's up direction there. Locked points are drawn in a different color.

Each point also shows a short reference arrow pointing along its “up” direction, so you can see how the profile is oriented before generating brushes.

Templates — the Swept Shape

The **template** is the cross-section that gets repeated along the curve. It can be a single brush, several brushes, or a whole group.

Control	What it does
Link	Uses the current selection as the template. Select a **group** to link by reference (later edits to the group update the spline), or select **one or more brushes** to link a snapshot of them.
Unlink	Detaches the template. The spline keeps its points but stops generating brushes until you link a template again.
Break	Duplicates the generated brushes as ordinary, editable brushes and then unlinks the template — in a single undo step. Use this when you are happy with the shape and want to hand-edit the result.

The template is swept along its **X axis**: the brush is copied and stretched to fit each segment of the curve while keeping its cross-section proportions. Generated brushes are **not selectable** on their own — edit the spline through its points, or press Break to release the brushes.

Curve Options

Control	What it does
Closed	Connects the last point back to the first, forming a loop. Brushes are generated on the closing segment too, and the seam is sealed. Great for rings, arches and racetracks.
Subdivisions	How finely the curve is sampled between control points. Higher values give a smoother curve (and more brushes); lower values are coarser and cheaper.
View Options	Standard TrenchBroom view options for the viewport.

Generated vertices are always snapped to the **grid (1 unit)**, and every swept cell is built from tetrahedra, so the geometry stays valid and watertight even on tight curves and twists.

Tips

- Splines are stored as **func_group** entities, so compilers merge the generated brushes into the world. They save and load with the map.
- Every change — adding, moving, rolling, closing, breaking — is a single **undoable** step.
- Keep the template compact along X for a tight cross-section; the sweep repeats it as many times as needed to cover the curve.
- If a twist “unwinds” over a segment on a closed loop, lock a point to control where the orientation is anchored.

TrenchBroom Spline Tool — user guide. Controls appear in the tool bar strip when the Spline Tool is active.